Abdelmoaty Mostafa Ibrahim

Process Safety & Asset Integrity | Natural Gas Engineering | GIS & Digital Transformation | Strategic Management
Oil & GAS

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PROFESSIONAL SUMMARY

Process Safety and Natural Gas Engineer Professional with over 18 years of experience spanning process safety compliance, natural gas network design, GIS, engineering data management, digital transformation, project execution and management systems. Experienced in compliance tracking, audit execution, CAPA follow-up, risk management, engineering design governance, spatial and hydraulic modeling, data governance, and multi-stakeholder infrastructure coordination. Combines engineering knowledge with digital solutions to improve data quality, design efficiency, operational control, and decision support.

PROCESS SAFETY & COMPLIANCE EXPERIENCE

- Current role includes compliance tracking, audit execution, CAPA follow-up, risk management and compliance reporting.
- Selected to support establishment of the company's Process Safety & Asset Integrity function in 2026.
- Completed Process Safety Job Shadowing at Rashpetco in 2026.
- Participated as Engineering Key User in the company's senior ISO certification committee and external audit discussions in 2017.
- Experience applying quality, environmental and occupational safety management requirements within engineering and project contexts.

NATURAL GAS ENGINEERING & DESIGN EXPERIENCE

- Over 18 years across natural gas network GIS, design, survey, data management, project execution and engineering management.
- Experience with distribution networks, transmission pipelines, commercial and industrial networks, and pressure-reduction infrastructure.
- Experience combining ArcGIS and Synergi Gas for spatial and hydraulic network modeling and design.
- Experience in survey, design, shop-drawing and as-built workflows and in final geospatial documentation of implemented infrastructure.

DIGITAL TRANSFORMATION & DATA GOVERNANCE

- Automated GIS/ERP/Synergi Gas workflows, reducing area-level design processing time by more than 95%.
- Led customer and gas-appliance data digitization and governance improvements using SQL Server.
- Supported SAP transformation as Engineering Key User through business-process analysis and requirements validation.
- Experience with data integration, spatial QA/QC, geodatabases, customer databases and engineering project databases.

PROJECT & STAKEHOLDER COORDINATION

- Coordinated work involving 16 contracting companies during Phase I Hayat Karima natural gas infrastructure activities.
- Coordinated with other utility stakeholders including electricity, potable water, wastewater and telecom to avoid conflicts and infrastructure damage.
- Worked under demanding delivery schedules and multi-party coordination requirements on major infrastructure and government projects.

CORE COMPETENCIES

- **Inherently Safer Design (ISD) and analysis:** Natural Gas Network Design through AutoCAD, Revit, GIS (ESRI) and Synergi Gas (DNV)
- **Enterprise Resource Planning (ERP) & Digital Transformation :** GIS, Synergi Gas , PLC, SCADA,SQL, SAP and Oracle
- **Process Safety Management (PSM):** HAZOP, LOPA, FMEA, BowTie Analysis, QRA.
- **Regulatory Compliance & Standards:** (API, NFPA, ASME, ISO 45001, ISO 14001, OSHA PSM, CCPS**)** Standards.
- **Risk Management & Emergency Response:** Crisis Management, Pre-Startup Safety Review (PSSR), ERP Development.
- **EHS Strategy & Operational Excellence:** Incident Investigation (Root Cause Analysis - RCA), Behavioral Safety (BBS).
- **Leadership & Audit Management:** Regulatory Audits, Contractor Safety Management, Cross-Functional Team Leadership.
- **GIS & Spatial Analytics:** ArcGIS Pro , ArcGIS Enterprise

SELECTED CAREER ACHIEVEMENTS

2009–2014 — GIS Automation & Digital Transformation

Developed and implemented custom ArcGIS Toolbox tools to automate spatial data QA/QC, validate customer and gas-appliance data from ERP, automatically link datasets using CRN (Client Reference Number), and generate Synergi Gas load files for downstream network design. Reduced area-level design processing time to less than 5% of the previous workflow, representing more than 95% reduction in processing time and effort.

2017 — ISO Certification Project

Selected as the Technical Affairs / Engineering Key User and representative on the company's senior ISO certification committee. Represented the Engineering function during the external audit and handled technical discussions with the audit team, contributing to successful achievement of ISO 9001:2015, ISO 14001:2015, and OHSAS 18001:2007 certification from the initial audit.

2017 — Government District, New Administrative Capital

Responsible for the engineering design of external and internal natural gas networks serving 36 major government buildings in the Government District, coordinating with multiple project stakeholders under demanding schedule constraints.

2018 — Customer & Gas Appliance Data Digitization and Governance

Led a company-wide project to improve field collection of customer and natural-gas-appliance data across all company regions. Supervised enhancement of the SQL Server application to strengthen data validation, cleansing, quality control, integration, and reporting, establishing a robust customer-data governance framework and enabling the transition from paper-based documentation to a controlled digital workflow.

2019–2022 — SAP Transformation Project

Selected as the Key User representing Technical Affairs / Engineering on the company's senior committee for the SAP transformation project, replacing the existing SQL Server-based system previously developed under my supervision. Conducted business-process analysis and requirements validation for Engineering, supporting successful implementation and Go-Live in 2022 on schedule despite exceptional operational challenges during the COVID-19 pandemic.

2022–2025 — Presidential Hayat Karima Initiative, Phase I

Nominated by the company to oversee Phase I natural gas infrastructure work across Sharqia, Qena, and Sohag. Supported spatial planning and PRS location planning; coordinated the work of 16 contracting companies and other utility stakeholders; and held responsibility for final review and approval of comprehensive ArcGIS Pro as-built geospatial databases documenting 5 Pressure Reduction Stations and networks serving 43 villages.

2025–2026 — Multi-Country Design Governance

Selected as Principal Engineer to establish and control natural gas network design parameters and design standards within ArcGIS Pro and Synergi Gas, aligning engineering criteria with applicable international codes and the regulatory, operational, and safety requirements of Saudi Arabia and the UAE.

2026–Present — Process Safety & Asset Integrity

Selected to support the establishment of the company's Process Safety & Asset Integrity function, building on experience in engineering, compliance, risk management, data governance, and natural gas infrastructure.

PROFESSIONAL EXPERIENCE

Process Safety Compliance Department Manager

(FEB. 2026 – Present)

- **Strategic Safety Governance**: Formulated and executed corporate PS strategies across multi-site natural gas distribution networks, lowering Total Recordable Incident Rate (TRIR).
- **Process Risk Assessments**: Coordinated for eliminating high-severity operational hazards for high-pressure gas stations and distribution facilities.
- **ISO System Integration**: Successfully led the transition and certification for ISO 45001 & ISO 14001 systems with zero non-conformities during external audits.
- **Emergency Preparedness**: Designed robust emergency response procedures and conducted major simulation exercises in coordination with local civil defense authorities.
- Implemented strict Contractor PS Compliance Programs, overseeing contractors during major EPC infrastructure expansions.
- Reduced workplace safety incidents through the introduction of a digital Incident Reporting & Root Cause Analysis system.

- Conducted comprehensive Pre-Startup Safety Reviews (PSSR) for newly commissioned gas pressure reduction stations.
- Managed day-to-day site PS operations, PTW (Permit to Work) systems, and job safety analyses (JSA).
- Delivered specialized Process Safety training programs to engineers and field technicians.

Director, Gas Network Design Dept.

(2025 – Feb. 2026)

- Review "survey, design, shop and as-built" drawings for natural gas networks to ensure compliance with design requirements.
- Spatial Analysis.
- Spatial & Hydraulic Modeling.
- Data Integration.
- Design gas distribution networks for cities.
- Design natural gas transmission pipelines between cities.
- Design natural gas networks for commercial and industrial facilities.

Director, Survey & Customer Data Dept.

(2024 – 2025)

- Land surveying.
- Field surveys for natural gas network design (People & Appliance Gas Consumption) (SAP PS Module).
- Data Integration between Arc GIS and SAP ERP.
- Spatial Data QA/QC.
- Spatial Analysis.
- Design and manage customer databases (SAP CS Module).
- Production of as-built maps supporting maintenance of utility networks (water, electricity, wastewater, and particularly natural gas in emergencies).

Director, Operations Dept.

(2023 – 2024)

- Development and management of customer service databases (SAP
- SD Module).
- Manage customer service department.

Director, Gas Network Project Execution Dept.

(2022 – 2023)

- Manage natural gas network projects according to PMP & ISO standards.
- Inspect completed gas networks and deliver them to EGAS (Egyptian
- Natural Gas Holding Company) in line with international standards.
- Design and manage material & operations databases (SAP PS Module).

Director, Survey & Customer Data Dept.

(2016 – 2022)

- Lead planning, technical development, and GIS teams, applying PMP & ISO standards.
- Prepare annual business plans to achieve competitive market share and scheduled targets.
- Provide strategic & technical studies and feasibility studies for natural gas network implementation.

- Design and manage project databases (SAP PS Module).

Engineering Consultants Group (ECG) – Saudi Arabia

(2014 – 2016)

- GIS Consultant – Roads Municipality, Holy Makkah.
- Projects:
    - Holy Makkah road maintenance projects consultancy.
    - Holy Makkah Urban development projects consultancy.
    - Six main road and bridges intersections development projects.

Cairo Gas Company – Petroleum Sector

(2009 – 2014)

GIS Engineer.

- Designed and developed GIS applications to integrate Synergi Gas with ArcGIS for natural gas distribution network modeling.
- Utilized geographic data for spatial analysis.
- Data Integration between Arc GIS, Synergi Gas and SQL Server.
- Spatial Data QA/QC.
- Designed and managed geodatabases (Arc GIS).
- Managed data collected by land and P&A survey teams.
- Managed customer databases (SQL Server).
- Produced "survey, design and as-built" drawings.

Egyptian Armed Forces

(2007 – 2008)

Computer & Information Systems Engineer.

TECHNICAL SKILLS

GIS: ArcGIS Pro, ArcGIS, QGIS, Geodatabase Management, Spatial Analysis, Spatial Data QA/QC, GIS Application Development, Web Mapping
Natural Gas Engineering: Natural Gas Network Design, Synergi Gas, Spatial & Hydraulic Modeling, Network Analysis, Route Analysis, Infrastructure Network Modeling
Data & Enterprise Systems: SAP ERP, SAP PS, SAP SD, SQL Server, SQL, PL/SQL, Oracle, Oracle Spatial, Data Integration, Data Governance
Engineering & CAD: AutoCAD, Land Survey, P&A Survey, Survey Mapping, Field Survey Mapping, As-Built Mapping, Production of Survey/Design/Shop/As-Built Drawings
Analytics & Business: Power BI, MS Project, MS Office, Urban Planning, Infrastructure Management, Utility Assets Management, Site Suitability Analysis, Environmental & Risk Analysis, 3D Modeling
Programming: C, C++, C#, VB, VBA, HTML5, CSS3

EDUCATION

B.Sc. Electrical Engineering — Computers, Control and Information Systems | Faculty of Engineering, Zagazig University | 2002–2007 | Grade: Good

CERTIFICATIONS & PROFESSIONAL TRAINING

- English Language Training – AUC:
    - Advanced Stage 401 - 2004
    - Advanced Stage 402 - 2005
- Oracle DBA Training — Oracle University
    - Oracle Database 10g : Introduction to SQL - 2005
    - Oracle Database 10g : Administration Work Shop I - 2005
    - Oracle Database 10g : Administration Work Shop II - 2006
- GIS Training — ESRI:
    - ArcGIS Desktop I - 2009
    - ArcGIS Desktop II - 2009
    - ArcGIS Desktop III - 2010
    - Building Geodatabases - 2010
    - Arc Objects with VBA - 2011
    - Arc Objects with .NET - 2011
- Quality, Environment & Safety Management Training — ISO 9001, ISO 14001, OHSAS 18001 (Upgraded to 45001) - 2017
- Project Management Training — 47 PDUs, Evorestte Company - 2018
    - Introduction to PMP – 12 PDUs
    - Project Management Professional – 35 PDUs
- Strategic Planning Training — OGS Company - 2018
- Personality Development Training — OGS Company - 2019
- Natural Gas Network Design and Analysis Training — Synergi Gas, DNV, 2025
- Natural Gas Network Advanced Design and Analysis Training — Synergi Gas, DNV, 2025

- Rashpetco Process Safety Training AS Job Shadowing - 2026

LANGUAGES

Arabic — Native | English — Excellent